

Project Initialization and Planning Phase

Date	5 th July 2026
Team ID	RAHMA HUSSEIN JUMA
Project Title	CREDIT CARD APPROVAL PREDICTION
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card approval using machine learning, boosting efficiency and accuracy. It tackles the inefficiencies of manual review, promising faster decisions, reduced risk of bad debt, and a better experience for both analysts and applicants. Key features include a machine learning-based credit risk model and near-instant decision-making through a web interface.

Project Overview	
Objective	The primary objective is to automate the credit card approval decision using machine learning, replacing slow manual review with a faster, more consistent, and more accurate assessment.
Scope	The project covers the full pipeline — collecting and cleaning applicant and credit history data, training and comparing four classification algorithms, and deploying the best-performing model through a web application for real-time predictions.
Problem Statement	
Description	Manually reviewing thousands of credit card applications every day is slow, inconsistent, and prone to human error, which affects both operational efficiency and the reliability of approval decisions.

Impact	Solving this results in faster application processing, more consistent risk assessment, and reduced exposure to bad debt from high-risk applicants who might otherwise slip through manual review.
Proposed Solution	
Approach	Employ machine learning to analyze applicant demographic, financial, and credit history data, predicting whether an applicant is likely to be a safe or risky borrower.
Key Features	- Machine learning-based credit risk assessment model, comparing Logistic Regression, Decision Tree, Random Forest, and XGBoost



	- Near-instant decision-making through a Flask web interface - Optional cloud deployment via IBM Watson Machine Learning for scalable hosting
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Standard CPU (no GPU required — classical ML models, not deep learning)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	~1 GB (dataset ~67 MB, model files a few hundred KB)
Software		

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, xgboost, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle dataset (rikdifos - Credit Card Approval Prediction): application_record.csv (438,510 rows, CSV) and credit_record.csv (1,048,575 rows, CSV), merged down to 36,457 labeled applicants